

TrES-3b

R-2582

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110218 · created 2026-08-27T01:05:14+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455610.96191 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.013
Transit depth (Rp/Rs)^2	3.85 +/- 0.49 [%]
Orbital Inclination (inc)	82.57 +/- 0.62
Airmass coefficient 1 (a1)	1.216 +/- 0.021
Airmass coefficient 2 (a2)	-0.1366 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.67 %
Best Comparison Star	None
Optimal Aperture	3.97
Optimal Annulus	13.71
Transit Duration (day)	0.0642 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.013 vs 0.16309 (deviation 2.53σ)
Duration fit vs published	0.0642 d vs 0.0567 d (deviation 1.57σ)
Mid-time O–C	1.5 min
Depth SNR	15.1
Residual scatter	1.67 %

Light curve

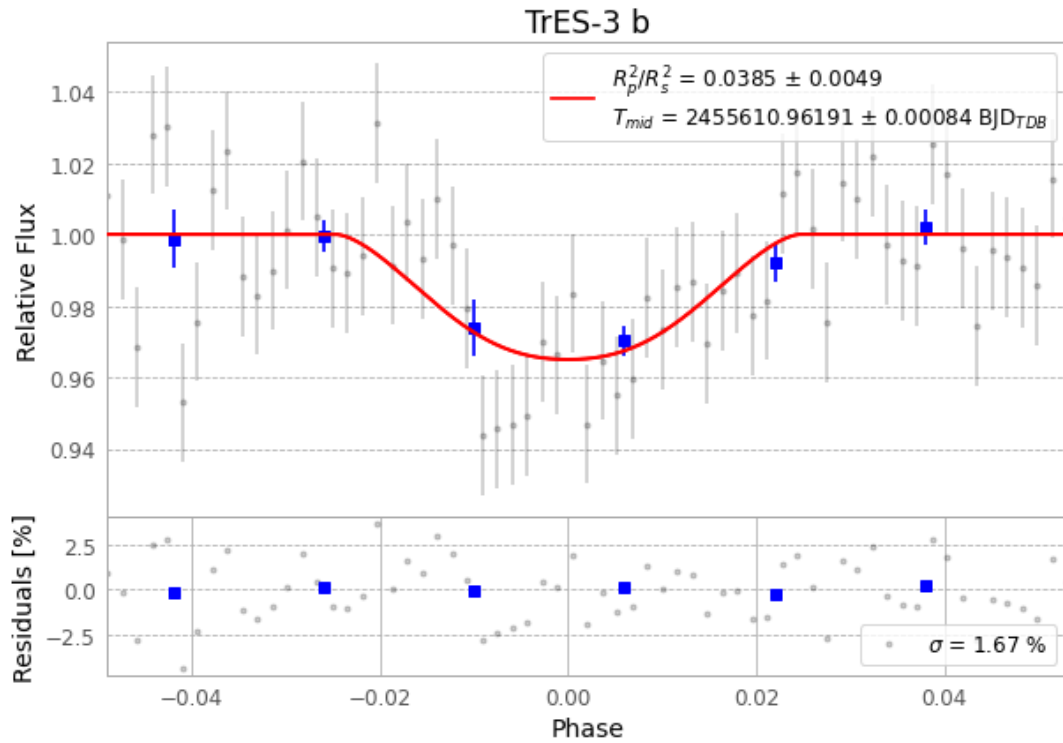


Figure 1 — FinalLightCurve_TrES-3 b_2011-02-18.png (SHA-256 30f3a5fab7fd2bca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 144], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 db4ad5a9ea23aedd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3110218093313.FITS → TRES-3110218124513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110218.log (6a3c071c517e...), FinalLightCurve_TrES-3 b_2011-02-18.png (30f3a5fab7fd...), FinalLightCurve_TrES-3 b_2011-02-18.csv (df833e97d862...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 01:05Z.